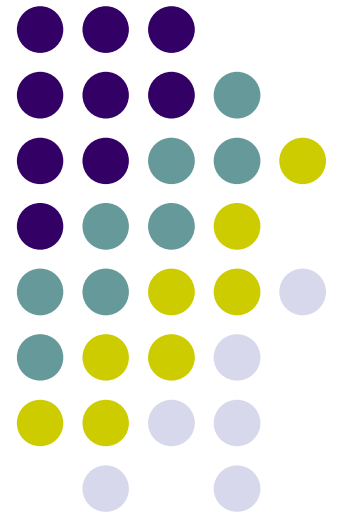


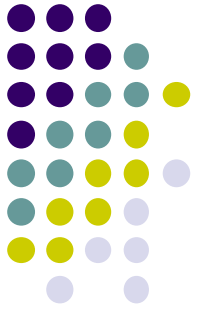
WIA1005 Network Technology Foundation

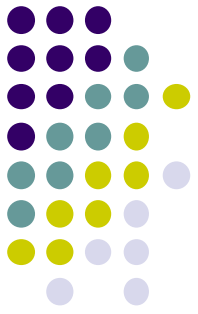
Chapter 10 DHCP



Contents

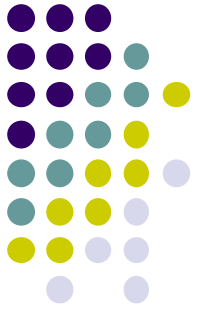
- DHCPv4
- DHCPv4 Configuration
- DHCPv6
- DHCP Attack





DHCPv4

- Dynamic Host Configuration Protocol (DHCPv4) assigns IPv4 addresses and other network configuration information dynamically.
- **Automatic Allocation** - DHCPv4 automatically assigns a static IPv4 address permanently to a device, selecting it from a pool of available addresses. There is no lease and the address is permanently assigned to the device.
- **Dynamic Allocation** - DHCPv4 dynamically assigns, or leases, an IPv4 address from a pool of addresses for a limited period of time chosen by the server, or until the client no longer needs the address.



DHCPv4

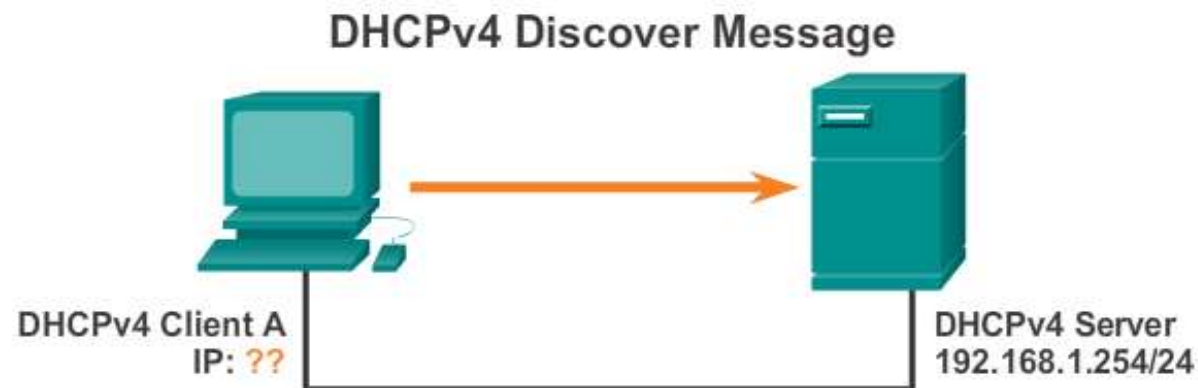
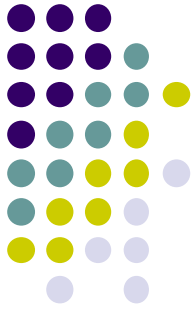
- **DHCP Discover**

- The DHCPDISCOVER message finds DHCPv4 servers on the network. It uses Layer 2 and Layer 3 broadcast addresses to communicate with the server.

- **DHCP Offer**

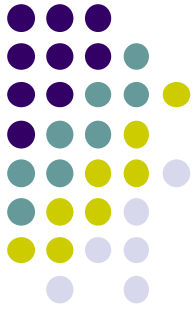
- When the DHCPv4 server receives a DHCPDISCOVER message, it reserves an available IPv4 address to lease to the client.
- The DHCPOFFER message is sent as a **unicast**, using the Layer 2 MAC address of the server as the source address and the Layer 2 MAC address of the client as the destination.

DHCPv4



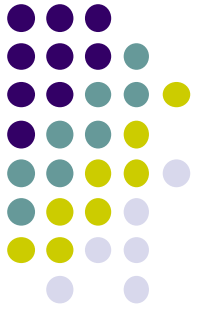
Ethernet Frame	IP	UDP	DHCPDISCOVER
SRC MAC: MAC A DST MAC: FF:FF:FF:FF:FF:FF	IP SRC: 0.0.0.0 IP DST: 255.255.255.255	UDP 67	CIADDR: 0.0.0.0 GIADDR: 0.0.0.0 Mask: 0.0.0.0 CHADDR: MAC A
MAC: Media Access Control Address CIADDR: Client IP Address GIADDR: Gateway IP Address CHADDR: Client Hardware Address			

DHCPv4



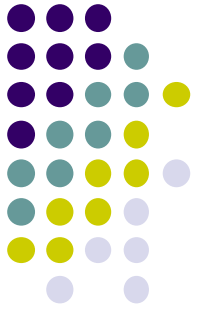
Ethernet Frame	IP	UDP	DHCP Reply
SRC MAC: MAC Serv DST MAC: MAC A	IP SRC: 192.168.1.254 IP DST: 192.168.1.10	UDP 68	CIADDR: 192.168.1.10 GIADDR: 0.0.0.0 Mask: 255.255.255.0 CHADDR: MAC A
MAC: Media Access Control Address CIADDR: Client IP Address GIADDR: Gateway IP Address CHADDR: Client Hardware Address			

DHCPv4



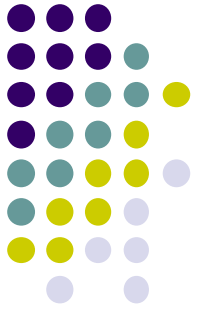
- **DHCP Request (DHCPREQUEST)**
 - When the client receives the DHCPOFFER from the server, it sends back a DHCPREQUEST message. The DHCPREQUEST serves as a binding acceptance notice to the selected server for the parameters it has offered and an implicit decline to any other servers that may have provided the client a binding offer.
 - The DHCPREQUEST message is sent in the form of a broadcast to inform this DHCPv4 server and any other DHCPv4 servers about the accepted offer.

DHCPv4

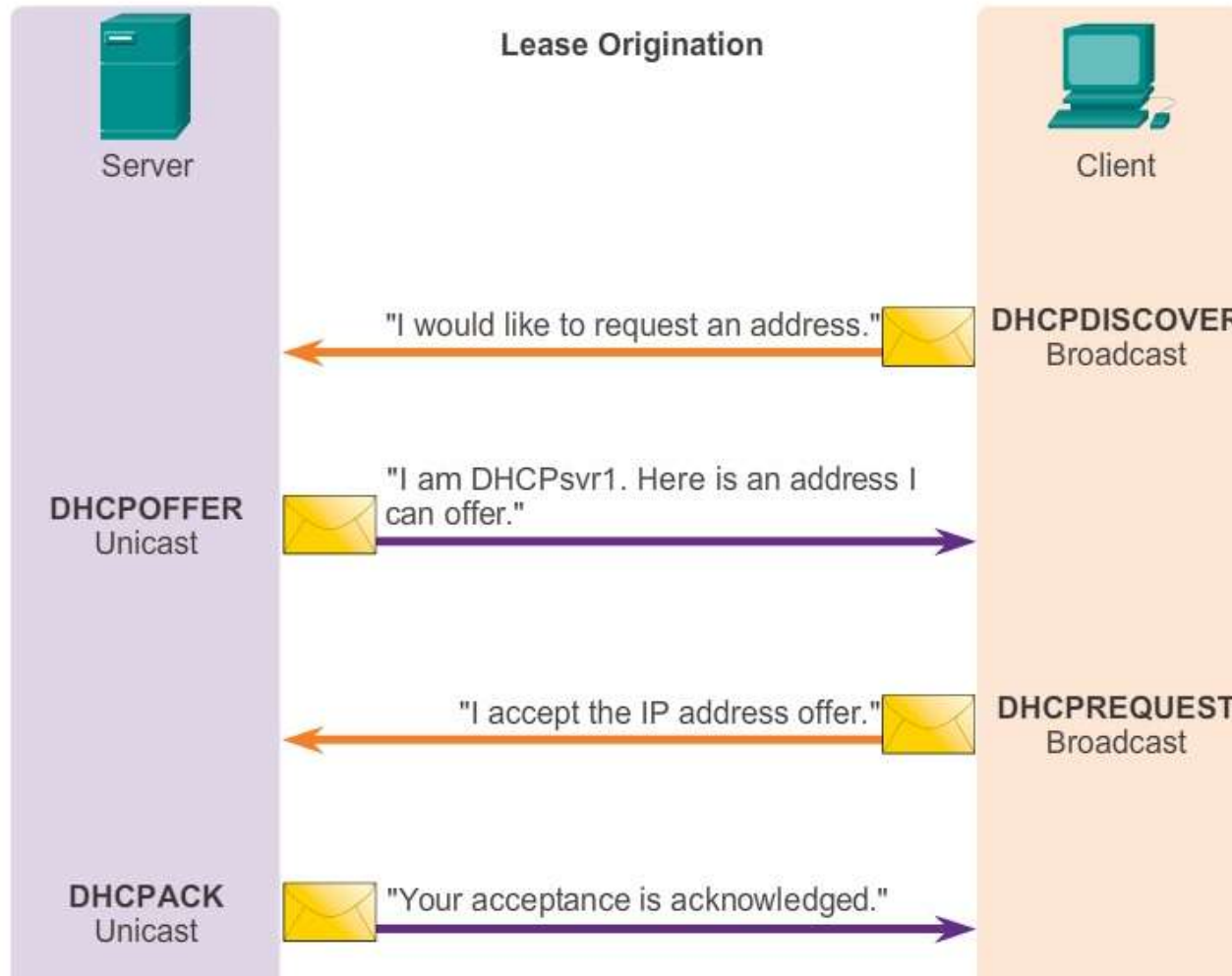


- **DHCP Acknowledgment (DHCPACK)**
 - On receiving the DHCPREQUEST message, the server verifies the lease information with an ICMP ping to that address to ensure it is not being used already, creates a new ARP entry for the client lease, and replies with a unicast DHCPACK message.
 - When the client receives the DHCPACK message, it logs the configuration information and performs an ARP lookup for the assigned address. If there is no reply to the ARP, the client knows that the IPv4 address is valid and starts using it as its own.

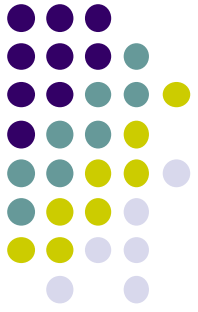
DHCPv4



DHCPv4 Operation



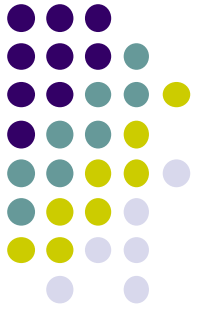
DHCPv4



- **Lease Renewal**

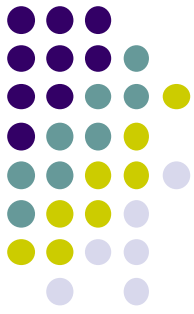
- When the lease has expired, the client sends a **DHCPREQUEST** message directly to the DHCPv4 server that originally offered the IPv4 address.
- If a DHCPACK is not received within a specified amount of time, the client broadcasts another DHCPREQUEST so that one of the other DHCPv4 servers can extend the lease.
- On receiving the DHCPREQUEST message, the server verifies the lease information by returning a DHCPACK.

DHCPv4



DHCPv4 Operation



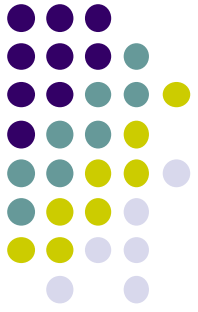


DHCPv4

- DHCPv4 messages are encapsulated within the UDP transport protocol.
- DHCPv4 messages sent from the client use UDP source port 68 and destination port 67.

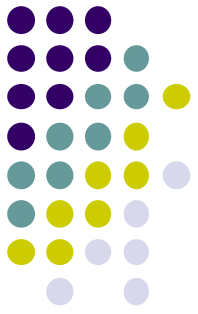
DHCPv4 Message Format

8	16	24	32
OP Code (1)	Hardware Type (1)	Hardware Address Length (1)	Hops (1)
Transaction Identifier			
Seconds - 2 bytes		Flags - 2 bytes	
Client IP Address (CIADDR) - 4 bytes			
Your IP Address (YIADDR) - 4 bytes			
Server IP Address (SIADDR) - 4 bytes			
Gateway IP Address (GIADDR) - 4 bytes			
Client Hardware Address (CHADDR) - 16 bytes			
Server Name (SNAME) - 64 bytes			
Boot Filename - 128 bytes			
DHCP Options - variable			



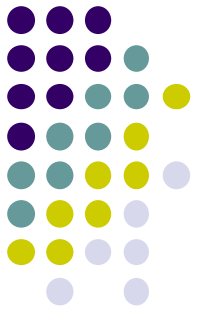
DHCPv4 Configuration

- A router can be configured to act as a DHCPv4 server.
- **Step 1. Excluding IPv4 Addresses**
 - Typically, some IPv4 addresses in a pool are assigned to network devices that require static address assignments.
 - Therefore, these IPv4 addresses should not be assigned to other devices.
- **Step 2. Configuring a DHCPv4 Pool**
 - Configuring a DHCPv4 server involves defining a pool of addresses to assign.



DHCPv4 Configuration

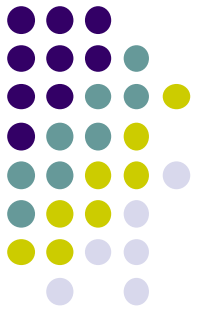
- **Step 3. Configuring Specific Tasks**
 - The tasks include the address pool and the gateway as well as some optional tasks such as dns server, domain name and lease time. The default lease time is one day.
- The DHCPv4 service is enabled, by default. To disable the service, use the **no service dhcp** command. Use the **service dhcp** command to re-enable the DHCPv4 service.
- The **show running-config | section dhcp** command displays the DHCPv4 commands configured on a router.



DHCPv4 Configuration

DHCPv4 Example

```
R1(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.9
R1(config)# ip dhcp excluded-address 192.168.10.254
R1(config)# ip dhcp pool LAN-POOL-1
R1(dhcp-config)# network 192.168.10.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.10.1
R1(dhcp-config)# dns-server 192.168.11.5
R1(dhcp-config)# domain-name example.com
R1(dhcp-config)# end
R1#
```

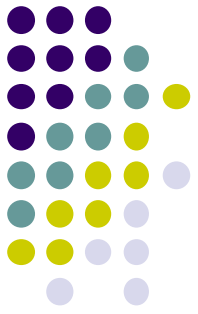


DHCPv4 Configuration

```
R1# show ip dhcp binding
```

```
Bindings from all pools not associated with VRF:
```

IP address	Client-ID/ Hardware address/ User name	Lease expiration	Type
192.168.10.10	0100.e018.5bdd.35	May 28 2013 01:06 PM	Automatic
192.168.11.10	0100.b0d0.d817.e6	May 28 2013 01:10 PM	Automatic

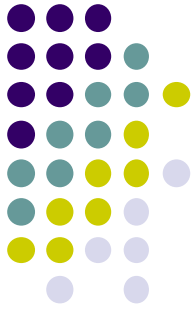


DHCPv4 Configuration

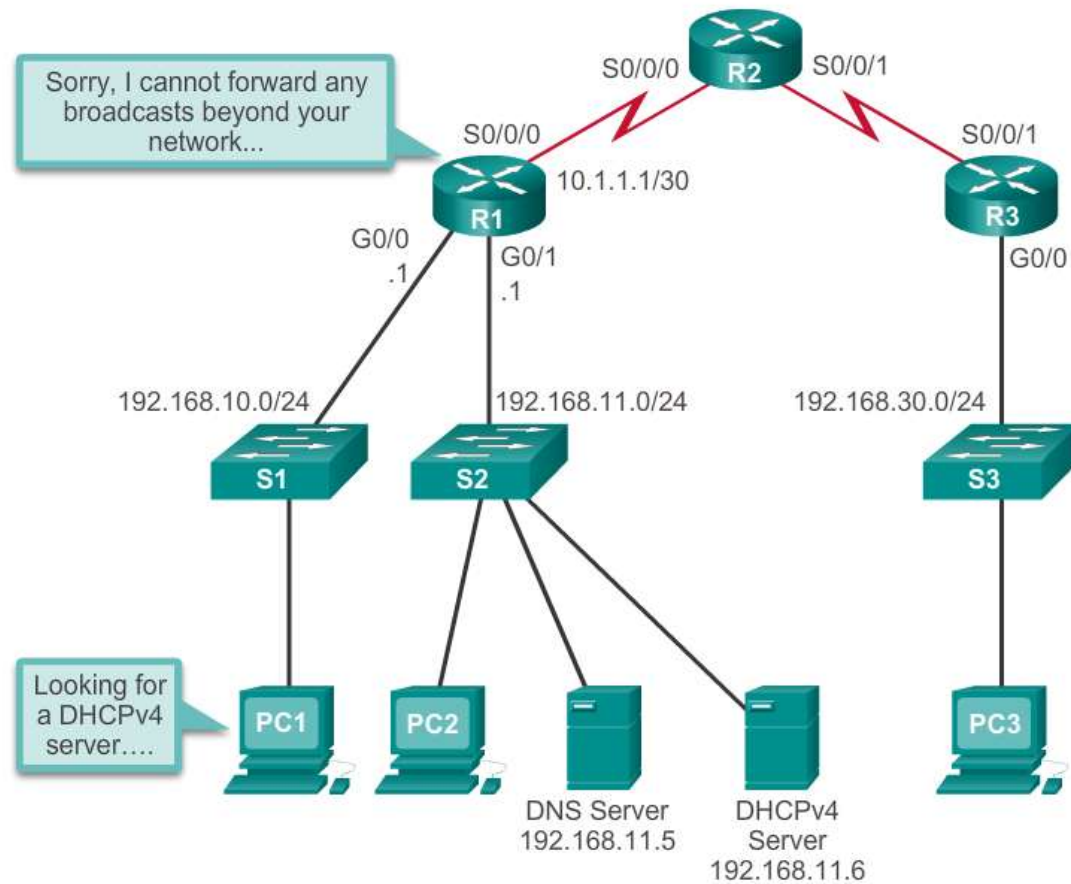
- **DHCP Relay**

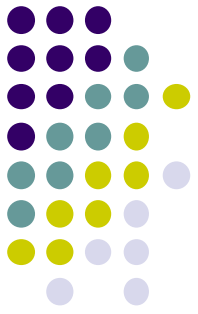
- In a complex hierarchical network, network clients are not typically on the same subnet as DHCP servers. In order to locate the servers and receive services, clients often use broadcast messages.
- DHCPv4 relay is a process to enable a router to forward DHCPv4 broadcasts to the DHCPv4 server.
- The **ip helper-address** command can be configured on the interface that receive the broadcast messages.

DHCPv4 Configuration



DHCPv4 Problems

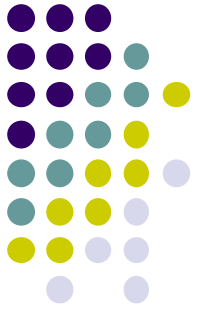




DHCPv4 Configuration

DHCPv4 Relay Commands

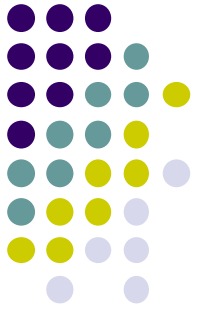
```
R1(config)# interface g0/0
R1(config-if)# ip helper-address 192.168.11.6
R1(config-if)# end
R1# show ip interface g0/0
GigabitEthernet0/0 is up, line protocol is up
  Internet address is 192.168.10.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is 192.168.11.6
<output omitted>
```



DHCPv4 Configuration

- By default, the **ip helper-address** command forwards the following eight UDP services:
 - Port 37: Time
 - Port 49: TACACS
 - Port 53: DNS
 - Port 67: DHCP/BOOTP client
 - Port 68: DHCP/BOOTP server
 - Port 69: TFTP
 - Port 137: NetBIOS name service
 - Port 138: NetBIOS datagram service

DHCPv4 Configuration

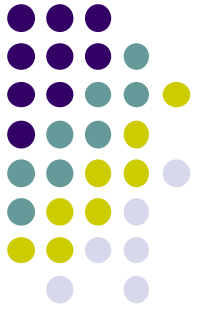


Configuring a Router as DHCP Client



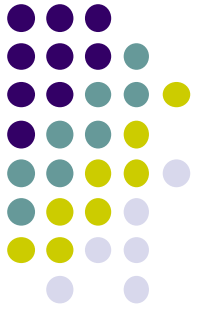
```
SOHO(config)# interface g0/1  
SOHO(config-if)# ip address dhcp  
SOHO(config-if)# no shutdown  
SOHO(config-if)#
```

DHCPv4 Configuration



Troubleshooting DHCPv4

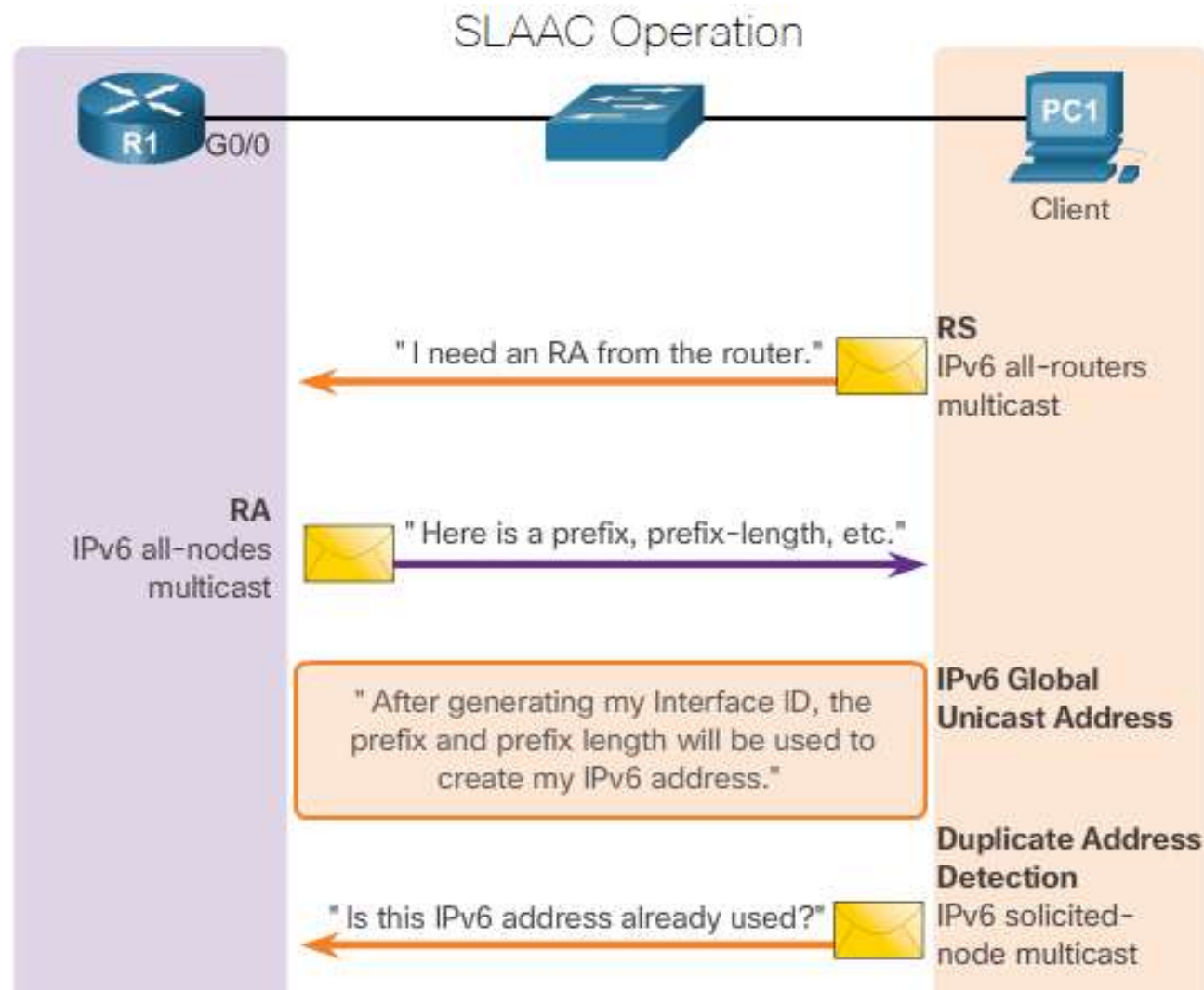
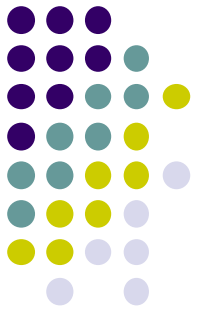
Troubleshooting Task 1:	Resolve address conflicts.
Troubleshooting Task 2:	Verify physical connectivity.
Troubleshooting Task 3:	Test with a static IPv4 address.
Troubleshooting Task 4:	Verify switch port configuration.
Troubleshooting Task 5:	Test from the same subnet or VLAN.



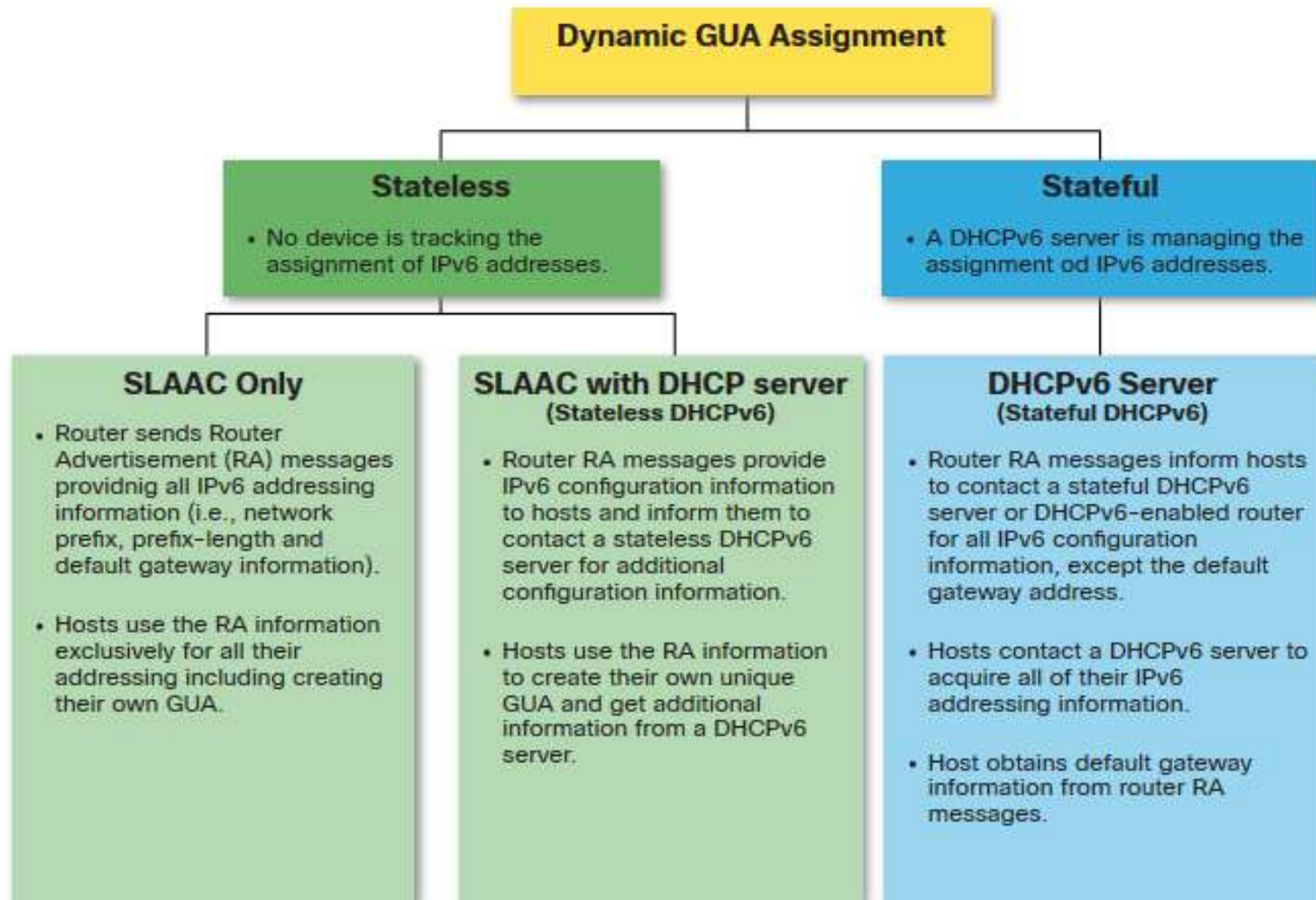
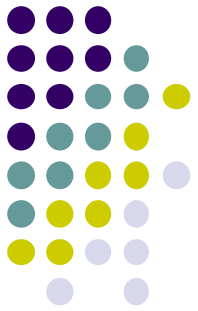
DHCPv6

- IPv6 global unicast addresses can be assigned dynamically:
 - Stateless Address Autoconfiguration (SLAAC) (**A flag**)
 - Dynamic Host Configuration Protocol for IPv6 (Stateful DHCPv6)
- ICMPv6 RA messages contain the Managed Address Configuration flag (**M flag**) and the Other Configuration flag (**O flag**) to determine how the client obtain its IPv6 addressing
- To set the flag, use **ipv6 nd other-config-flag** command and **ipv6 nd managed-config-flag** command.

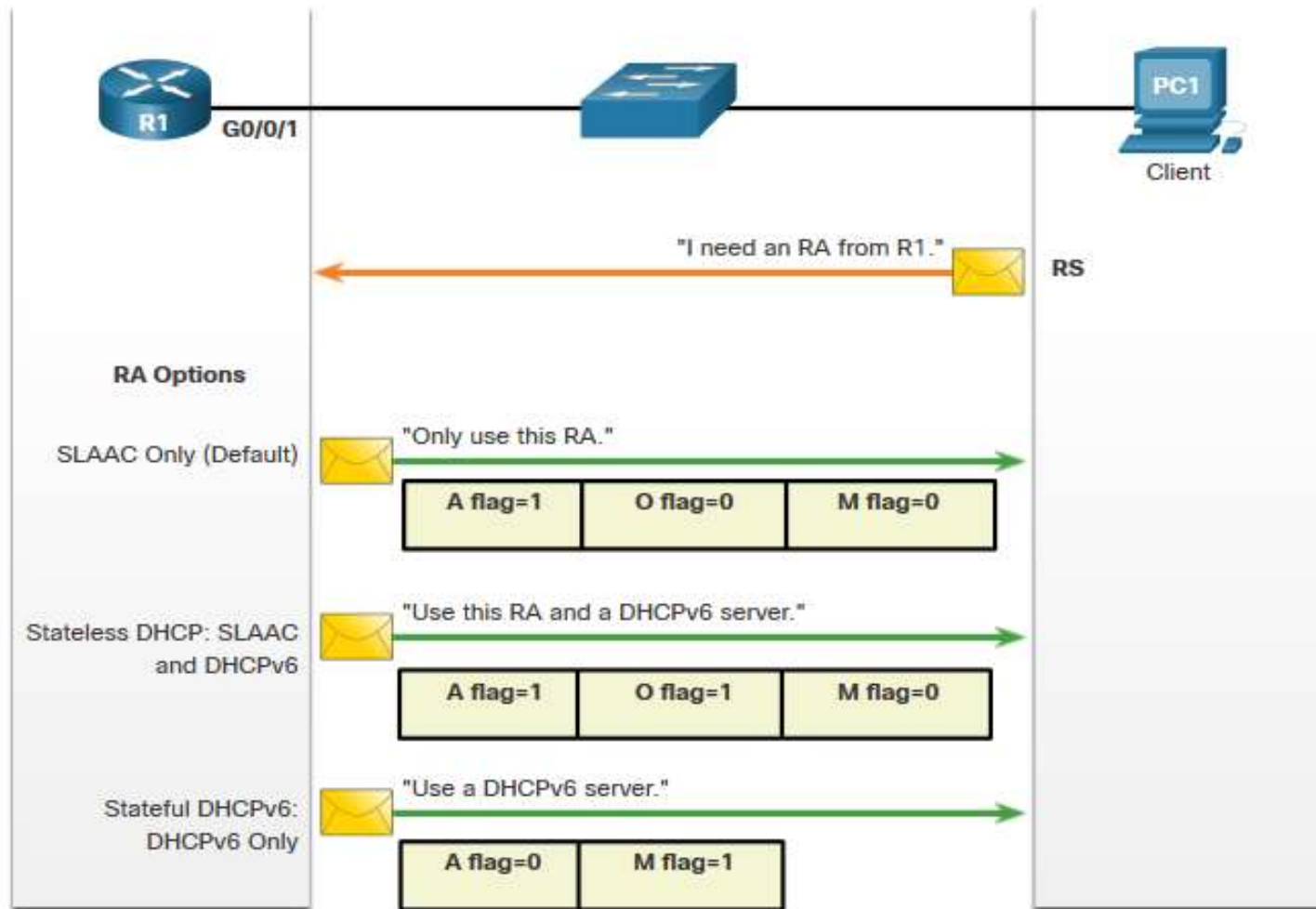
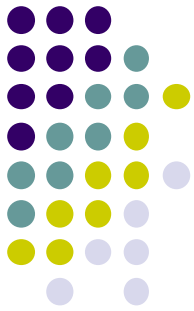
DHCPv6



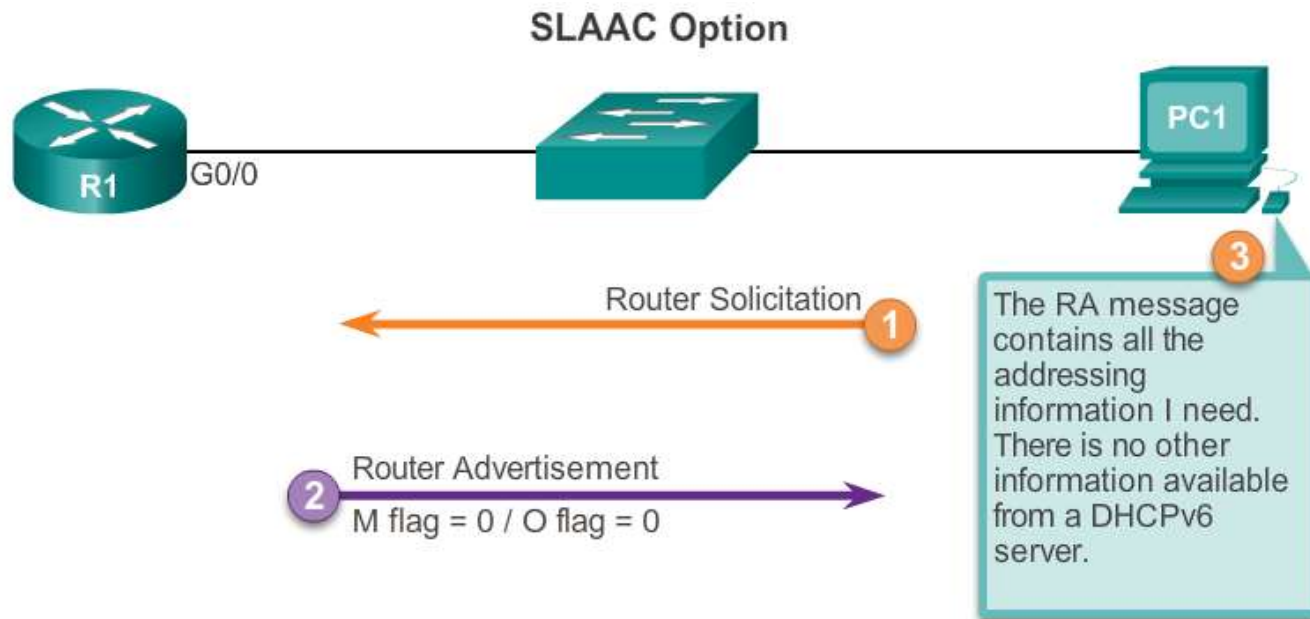
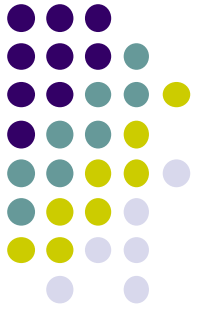
DHCPv6



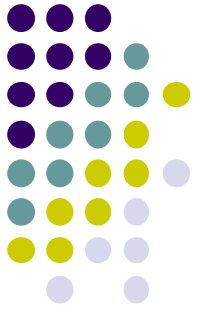
DHCPv6



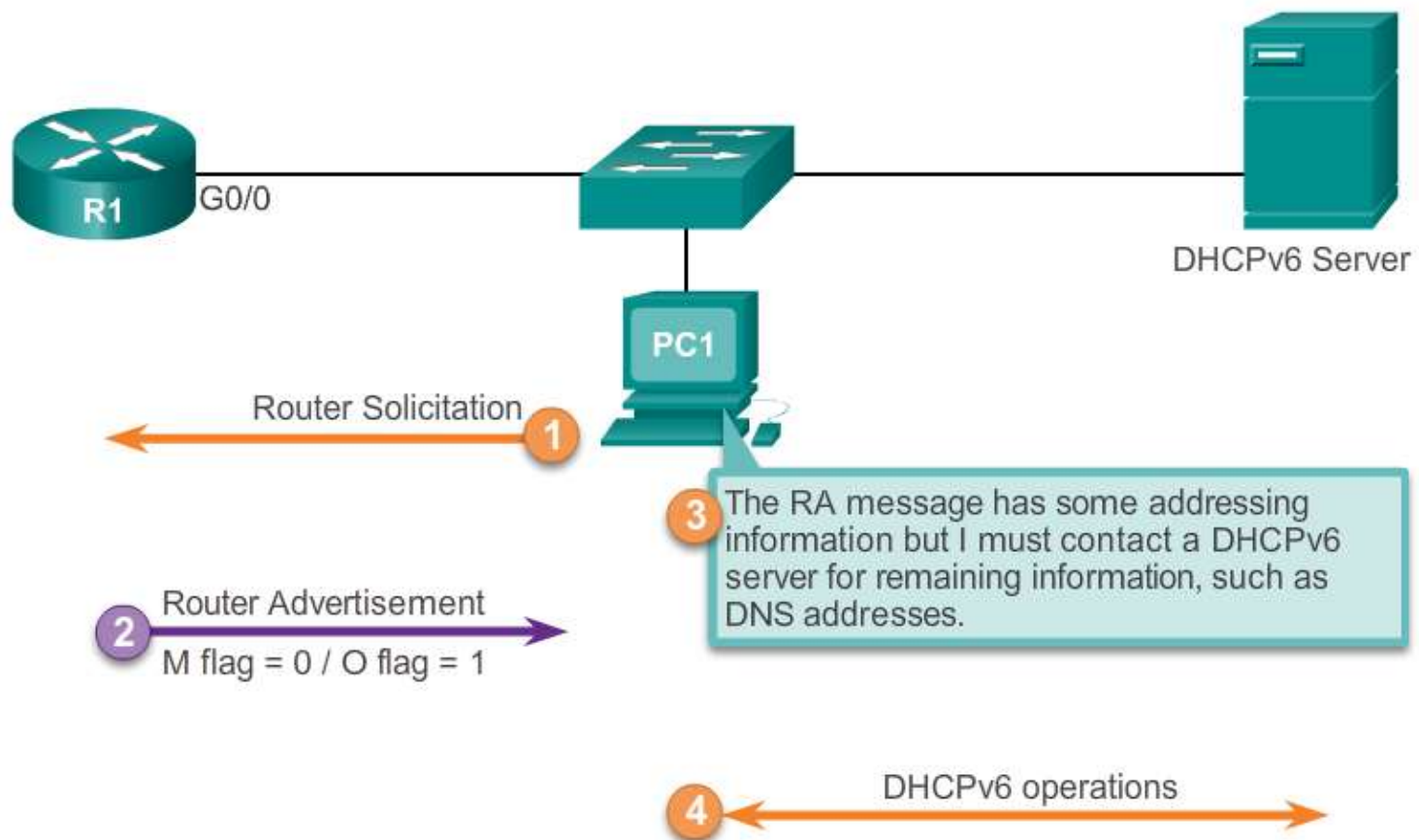
DHCPv6



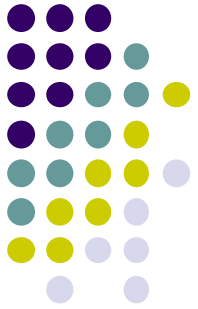
DHCPv6



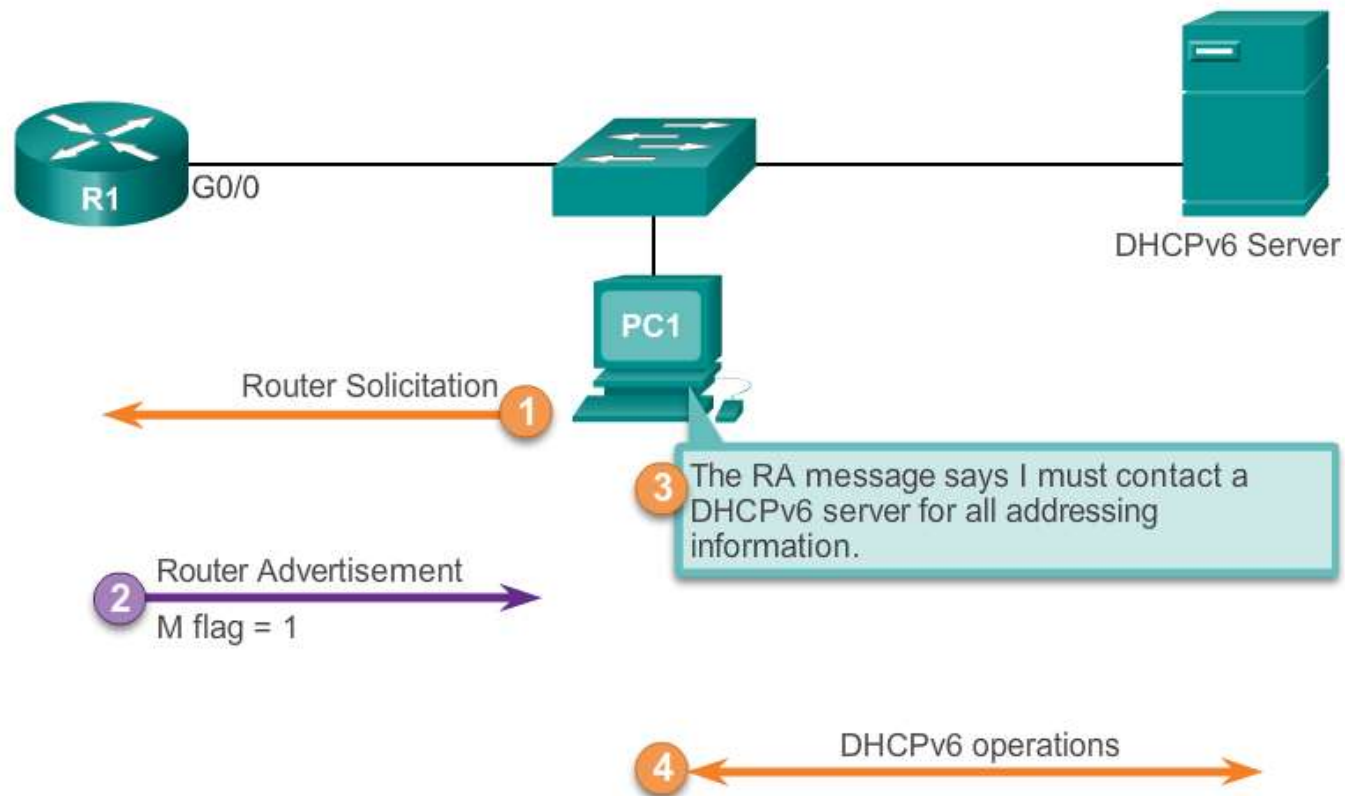
Stateless DHCPv6 Option

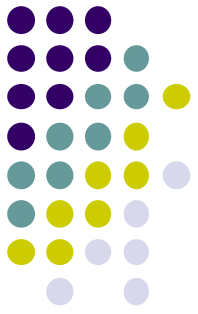


DHCPv6



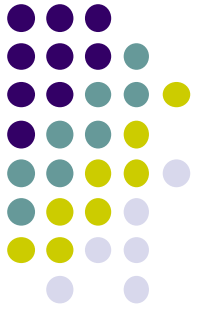
Stateful DHCPv6 Option





DHCPv6

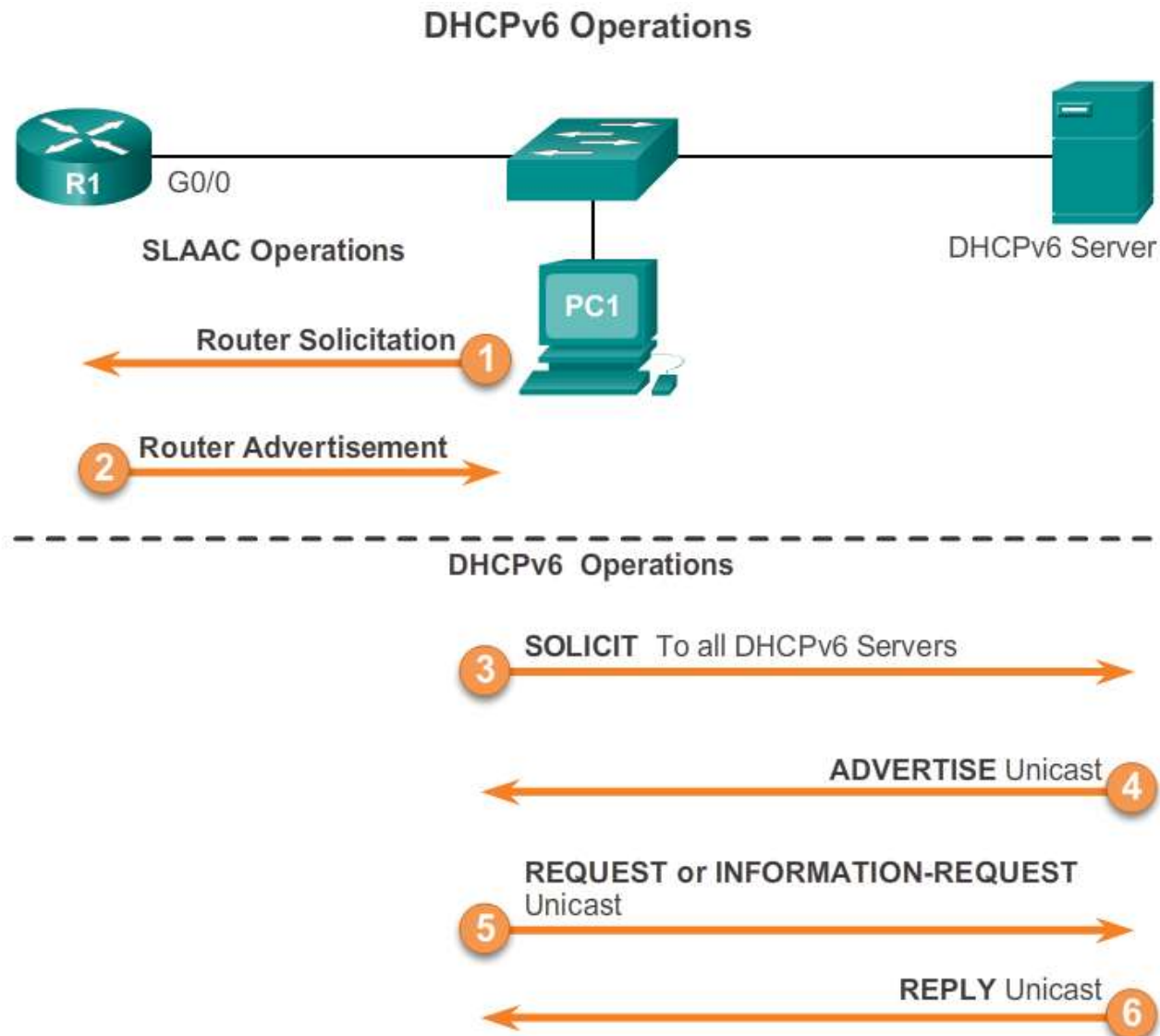
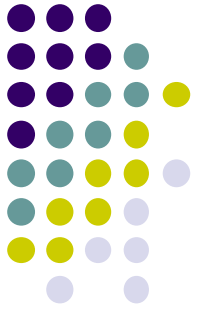
- DHCPv6 messages are sent over UDP. DHCPv6 messages from the server to the client use UDP destination port 546. The client sends DHCPv6 messages to the server using UDP destination port 547.
- First, the client sends a **DHCPv6 SOLICIT** message to the reserved IPv6 multicast all-DHCPv6-servers address **FF02::1:2**. This multicast address has link-local scope, which means routers do not forward the messages to other networks.
- One or more DHCPv6 servers respond with a **DHCPv6 ADVERTISE** message. The ADVERTISE message informs the DHCPv6 client that the server is available for DHCPv6 service.

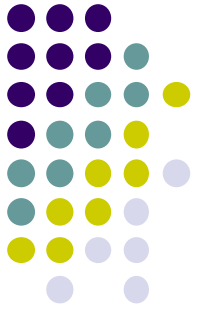


DHCPv6

- The **Stateless DHCPv6 client** responds with a **DHCPv6 INFORMATION-REQUEST** message to the DHCPv6 server requesting only configuration parameters, such as DNS server address.
- The **Stateful DHCPv6 client** responds with a **DHCPv6 REQUEST** message to the server to obtain an IPv6 address and all other configuration parameters from the server.
- The server sends a **DHCPv6 REPLY** to the client containing the information requested.

DHCPv6





DHCPv6 Configuration

Configuring Stateless DHCPv6 on a Router

Step 1: Enable IPv6 Routing

```
Router(config)# ipv6 unicast-routing
```

Step 2: Configure a DHCPv6 Pool

```
Router(config)# ipv6 dhcp pool pool-name  
Router(config-dhcpv6)#
```

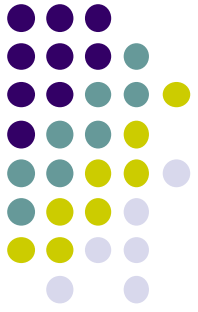
Step 3: Configure Pool Parameters

```
Router(config-dhcpv6)# dns-server dns-server-address  
Router(config-dhcpv6)# domain-name domain-name
```

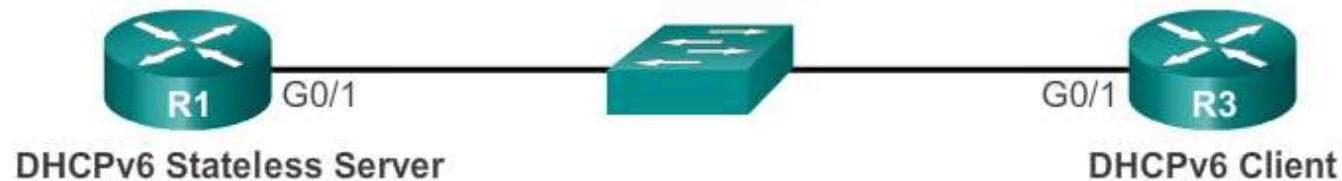
Step 4: Configure the DHCPv6 Interface

```
Router(config)# interface type number  
Router(config-if)# ipv6 dhcp server pool-name  
Router(config-if)# ipv6 nd other-config-flag
```

DHCPv6 Configuration

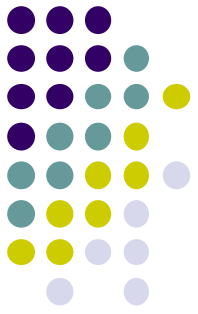


Configuring Router R1 as Stateless DHCPv6 Server

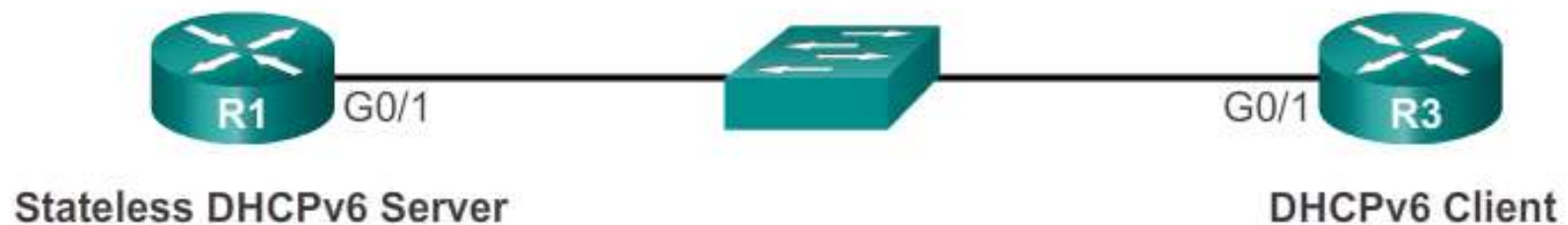


```
R1(config)# ipv6 unicast-routing
R1(config)# ipv6 dhcp pool IPV6-STATELESS
R1(config-dhcpv6)# dns-server 2001:db8:cafe:aaaa::5
R1(config-dhcpv6)# domain-name example.com
R1(config-dhcpv6)# exit
R1(config)# interface g0/1
R1(config-if)# ipv6 address 2001:db8:cafe:1::1/64
R1(config-if)# ipv6 dhcp server IPV6-STATELESS
R1(config-if)# ipv6 nd other-config-flag
```

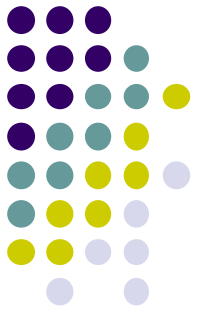
DHCPv6 Configuration



Configuring a Router as Stateless DHCPv6 Client



```
R3(config)# interface g0/1  
R3(config-if)# ipv6 enable  
R3(config-if)# ipv6 address autoconfig
```



DHCPv6 Configuration

Configuring a Stateful DHCPv6 Router

Step 1: Enable IPv6 Routing

```
Router(config)# ipv6 unicast-routing
```

Step 2: Configure a DHCPv6 Pool

```
Router(config)# ipv6 dhcp pool pool-name  
Router(config-dhcpv6)#
```

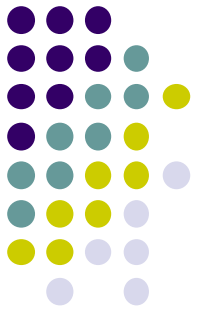
Step 3: Configure Pool Parameters

```
Router(config-dhcpv6)# address prefix/length [lifetime  
                                valid-lifetime preferred-lifetime  
                                | infinite]  
Router(config-dhcpv6)# dns-server dns-server-address  
Router(config-dhcpv6)# domain-name domain-name
```

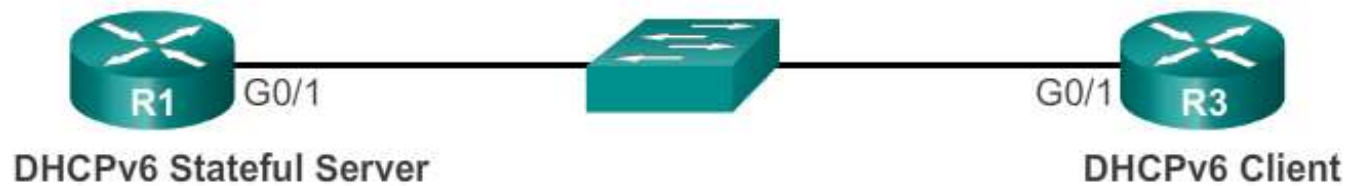
Step 4: Configure the DHCPv6 Interface

```
Router(config)# interface type number  
Router(config-if)# ipv6 dhcp server pool-name  
Router(config-if)# ipv6 nd managed-config-flag
```

DHCPv6 Configuration

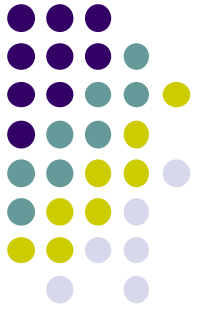


Configuring Router R1 as Stateful DHCPv6 Server

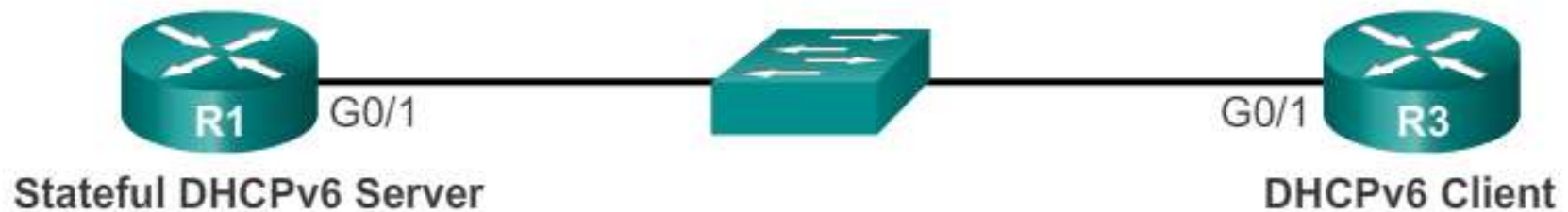


```
R1(config)# ipv6 unicast-routing
R1(config)# ipv6 dhcp pool IPV6-STATEFUL
R1(config-dhcpv6)# address prefix 2001:DB8:CAFE:1::/64
                    lifetime infinite
R1(config-dhcpv6)# dns-server 2001:db8:cafe:aaaa::5
R1(config-dhcpv6)# domain-name example.com
R1(config-dhcpv6)# exit
R1(config)# interface g0/1
R1(config-if)# ipv6 address 2001:db8:cafe:1::1/64
R1(config-if)# ipv6 dhcp server IPV6-STATEFUL
R1(config-if)# ipv6 nd managed-config-flag
```

DHCPv6 Configuration

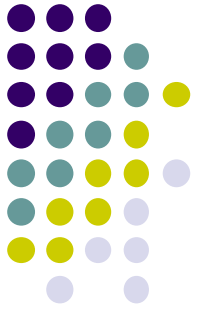


Configuring a Router as Stateful DHCPv6 Client



```
R3(config)# interface g0/1
R3(config-if)# ipv6 enable
R3(config-if)# ipv6 address dhcp
R3(config-if)#
```


DHCPv6 Configuration



```
R1# show ipv6 dhcp pool
```

```
DHCPv6 pool: IPV6-STATEFUL
```

```
Address allocation prefix: 2001:DB8:CAFE:1::/64 valid  
4294967295 preferred 4294967295 (1 in use, 0 conflicts)
```

```
DNS server: 2001:DB8:CAFE:AAAA::5
```

```
Domain name: example.com
```

```
Active clients: 1
```

```
R1# show ipv6 dhcp binding
```

```
Client: FE80::32F7:DFE:FE25:2DE1
```

```
DUID: 0003000130F70D252DE0
```

```
Username : unassigned
```

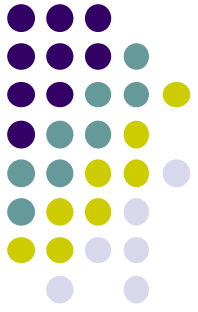
```
IA NA: IA ID 0x00040001, T1 43200, T2 69120
```

```
Address: 2001:DB8:CAFE:1:5844:47B2:2603:C171
```

```
preferred lifetime INFINITY, , valid lifetime
```

```
INFINITY,
```

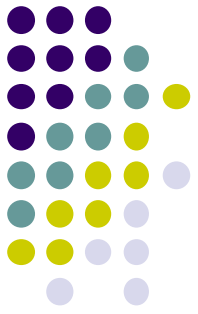
```
R1#
```



DHCPv6

- If the DHCPv6 server is located on a different network than the client, then the IPv6 router can be configured as a DHCPv6 relay agent.
- A DHCPv6 relay agent is configured using the **ipv6 dhcp relay destination** command. This command is configured on the interface facing the DHCPv6 client using the address of the DHCPv6 server as the destination.

```
R1(config)# interface g0/0
R1(config-if)# ipv6 dhcp relay destination 2001:db8:cafe:1::6
R1(config-if)# end
R1# show ipv6 dhcp interface g0/0
GigabitEthernet0/0 is in relay mode
Relay destinations:
  2001:DB8:CAFE:1::6
```

DHCP Attack

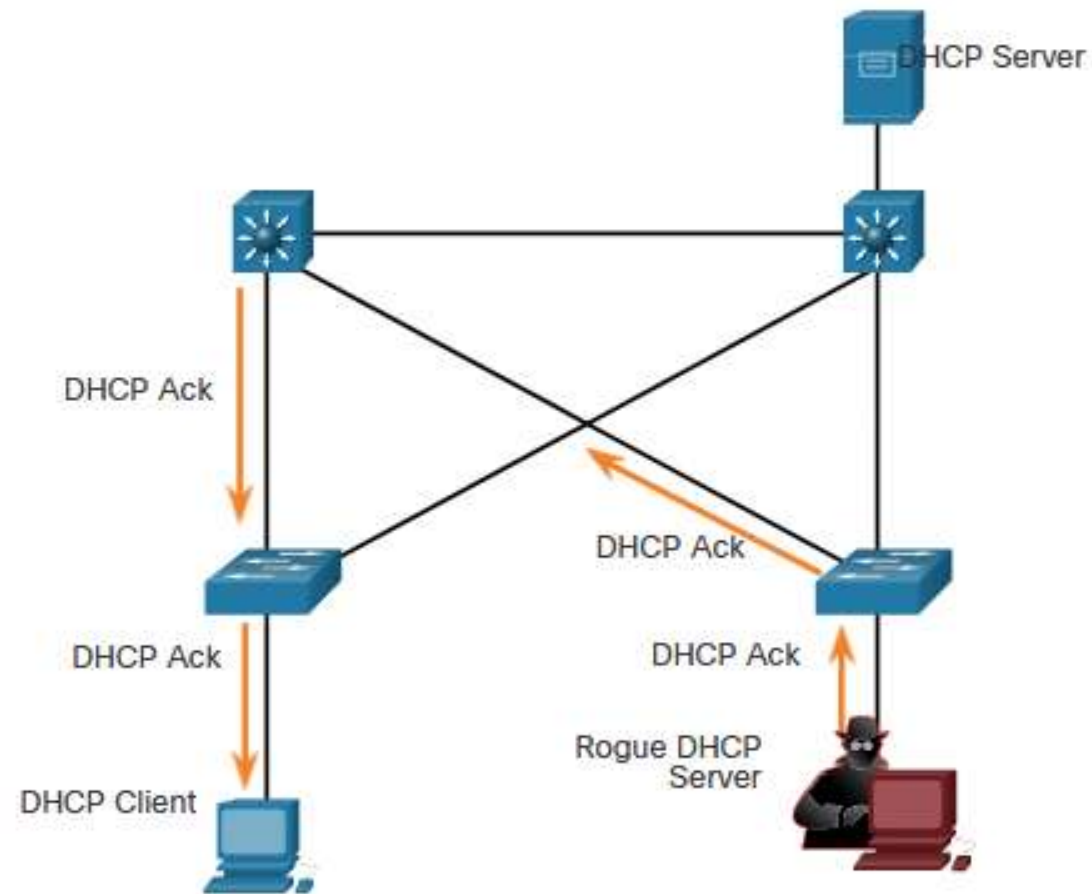
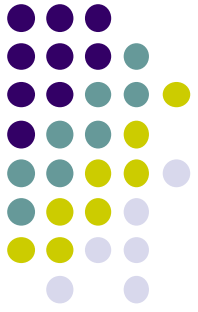
- **DHCP starvation attacks**

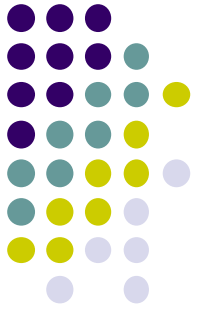
- An attacker **floods the DHCP server with DHCP requests** to use up all the available IP addresses that the DHCP server can issue. (Gobbler)
- After these IP addresses are issued, the server cannot issue any more addresses, and this situation produces a denial-of-service (DoS) attack as new clients cannot obtain network access.

- **DHCP spoofing attacks**

- An attacker configures a fake DHCP server on the network to issue DHCP addresses to clients.
- The attack is to force the clients to use machine under the control of the attacker, as their default gateway.

DHCP Attack

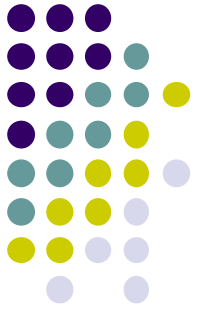




DHCP Attack

- **DHCP Snooping**

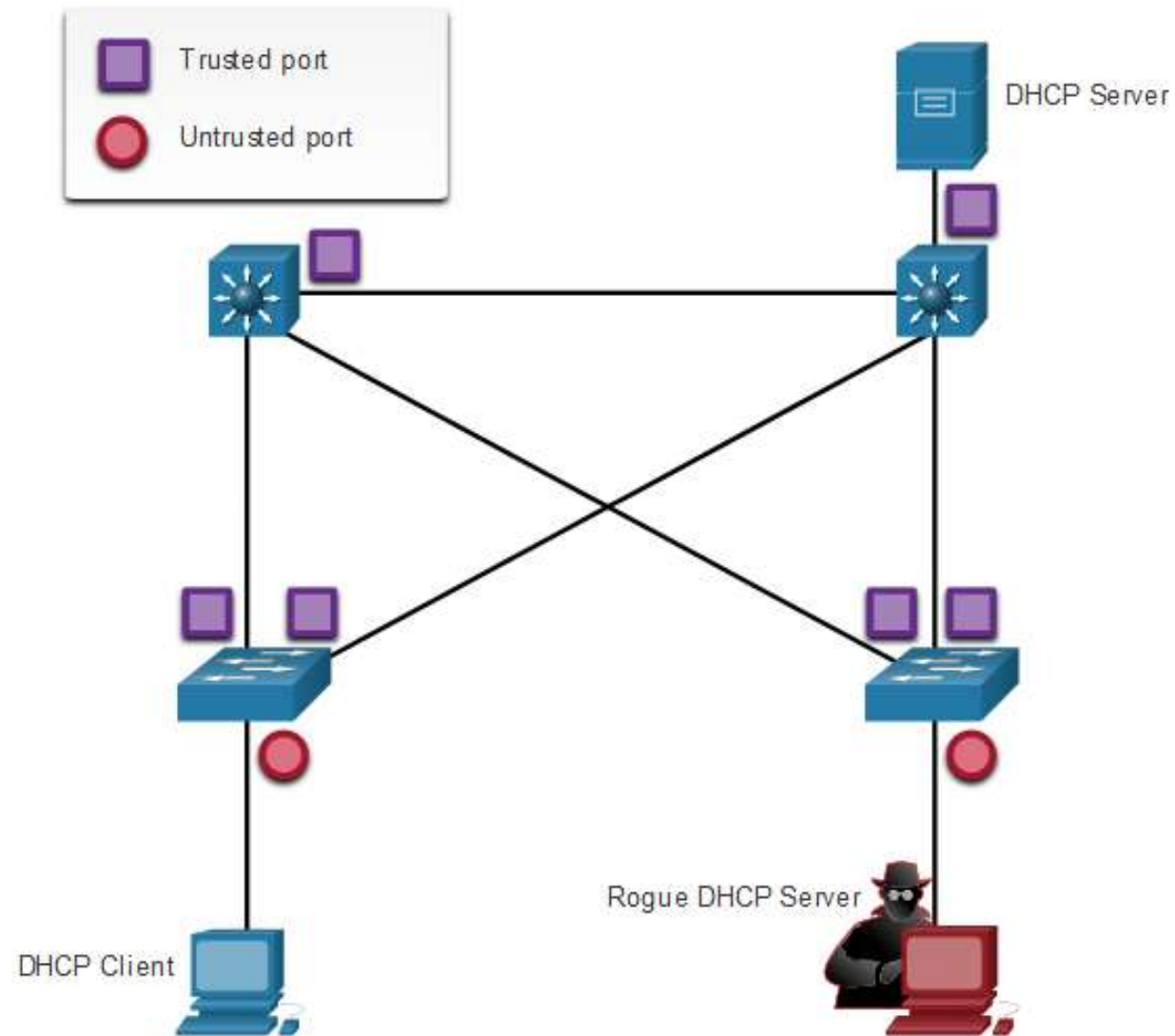
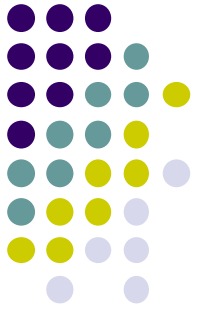
- DHCP snooping determines whether DHCP messages are from an administratively configured trusted or untrusted source.
- Devices under your administrative control, such as switches, routers, and servers, are trusted sources.
- In addition, all access ports are generally treated as untrusted sources.
- Trusted ports can source all DHCP messages; untrusted ports can source requests only.

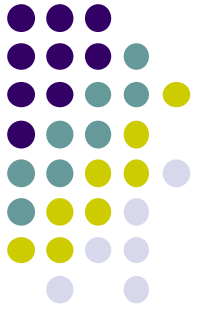


DHCP Attack

- All interfaces are treated as **untrusted by default**. Trusted interfaces are typically trunk links and ports directly connected to a legitimate DHCP server.
- If a rogue device on an untrusted port attempts to send a DHCP response packet into the network, the port is shut down.
- A DHCP table is built that includes the source MAC address of a device on an untrusted port and the IP address assigned by the DHCP server to that device.
- The MAC address and IP address are bound together. Therefore, this table is called the DHCP snooping binding table.

DHCP Attack

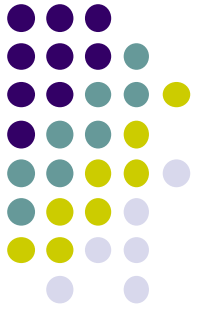




DHCP Attack

- Step 1. Enable DHCP snooping by using the ip dhcp snooping
- Step 2. On trusted ports, use the ip dhcp snooping trust
- Step 3: Limit the number of DHCP discovery messages that can be received per second on untrusted ports by using the ip dhcp snooping limit rate. The rate limit is set to **number of packets** per second.

DHCP Attack



DHCP Snooping Configuration

```
S1(config)# ip dhcp snooping
S1(config)# ip dhcp snooping vlan 10,20
S1(config)# interface fastethernet 0/1
S1(config-if)# ip dhcp snooping trust
S1(config)# interface fastethernet 0/2
S1(config-if)# ip dhcp snooping limit rate 5
```

